COMPUTER VISION

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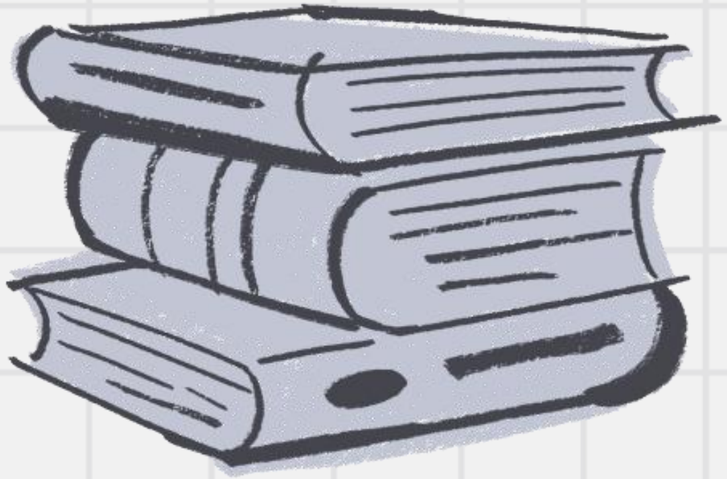
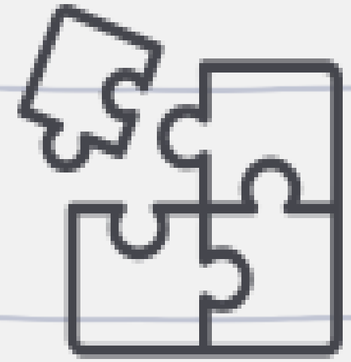


01



Introduction

Introduction

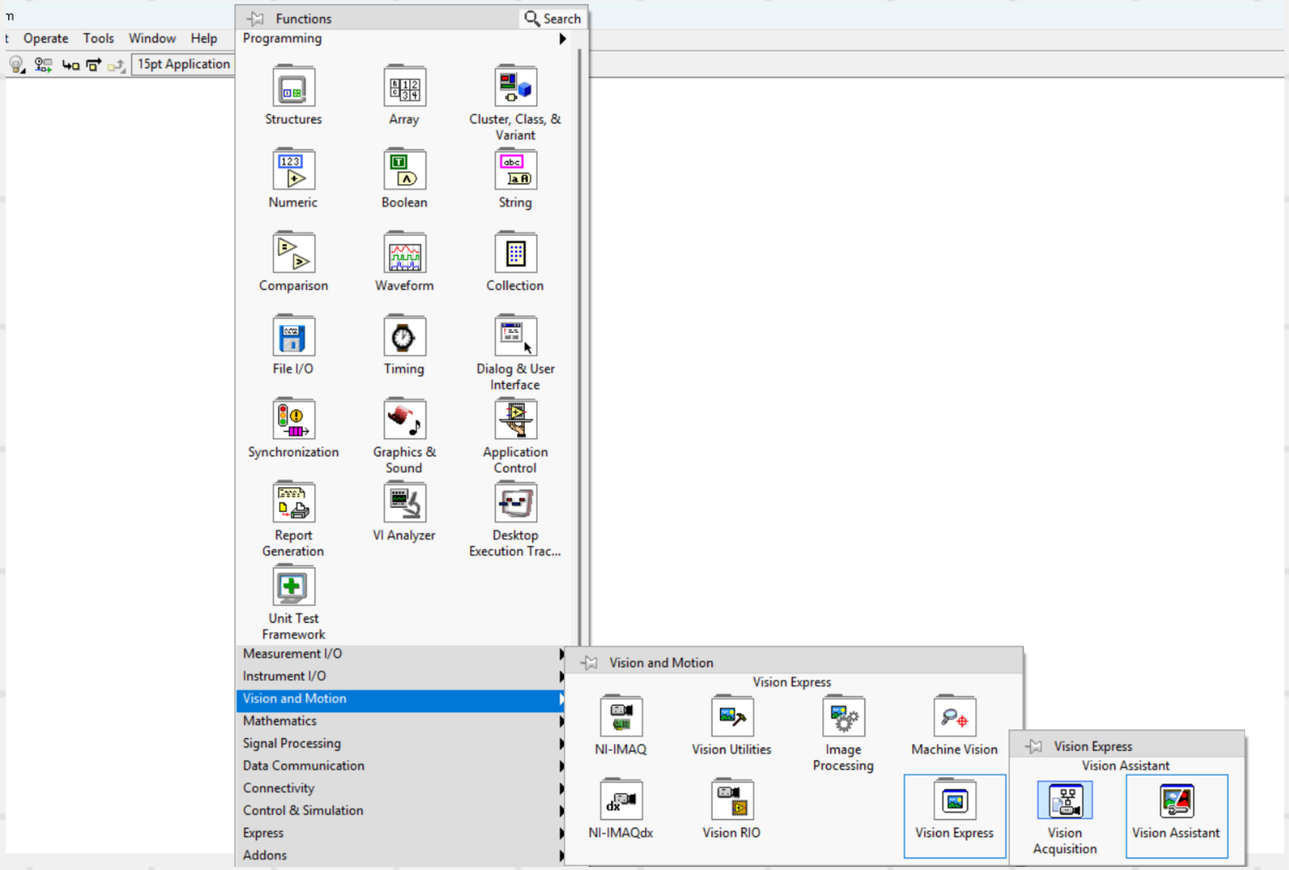
Now that we have most of the foundational knowledge, we’re almost ready to build our project.
However, there are still a few technical steps we need to take.
The first of these is using a real-time camera to capture a live video stream, which is essential for performing object tracking.
In this session, we’ll walk through that process step by step.

02

Real-time Camera

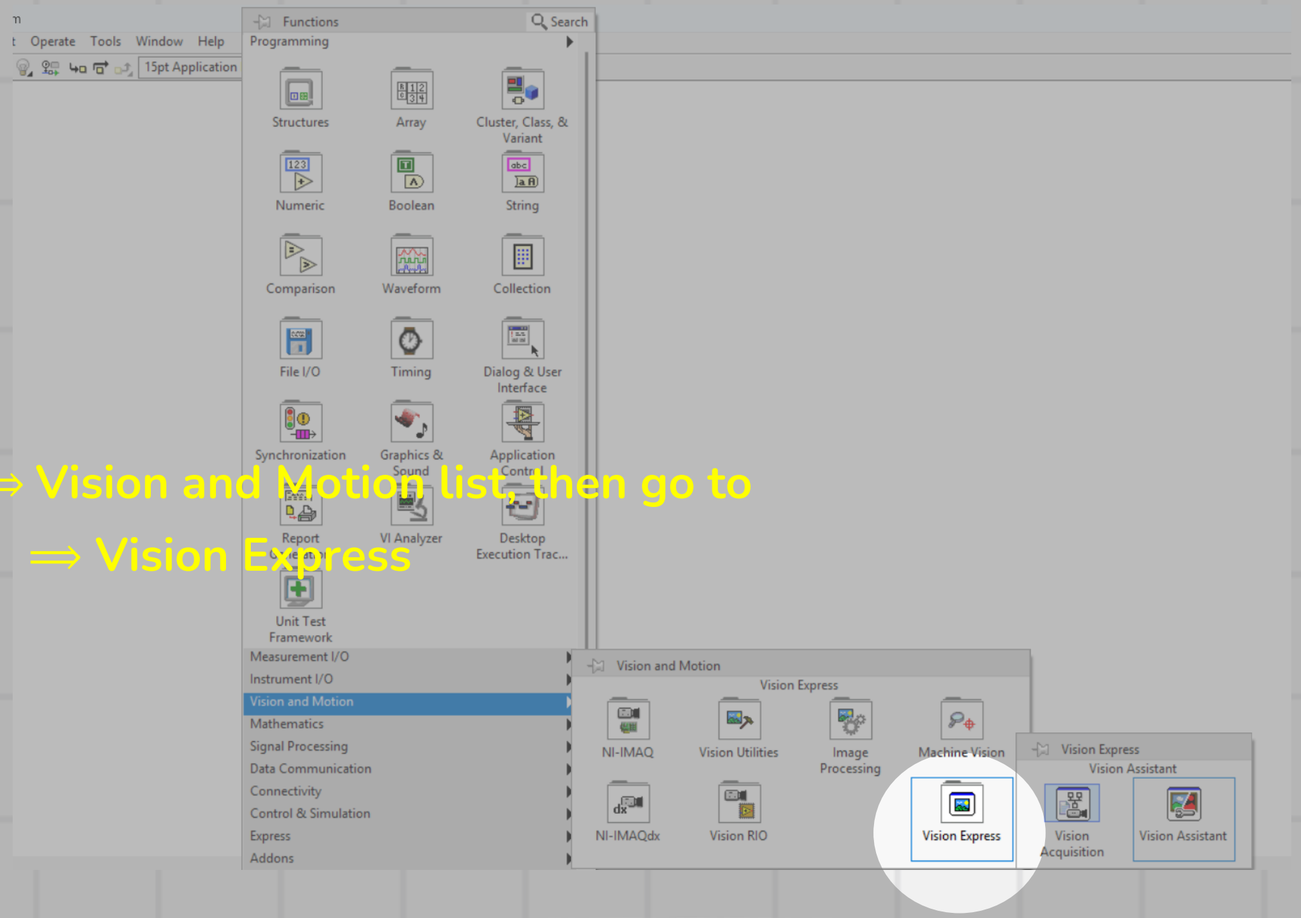
Real-time Camera

Simplest way to do this is by depending on vision acquisition app.



Real-time Camera

Simplest way to do this is by depending on vision acquisition app.

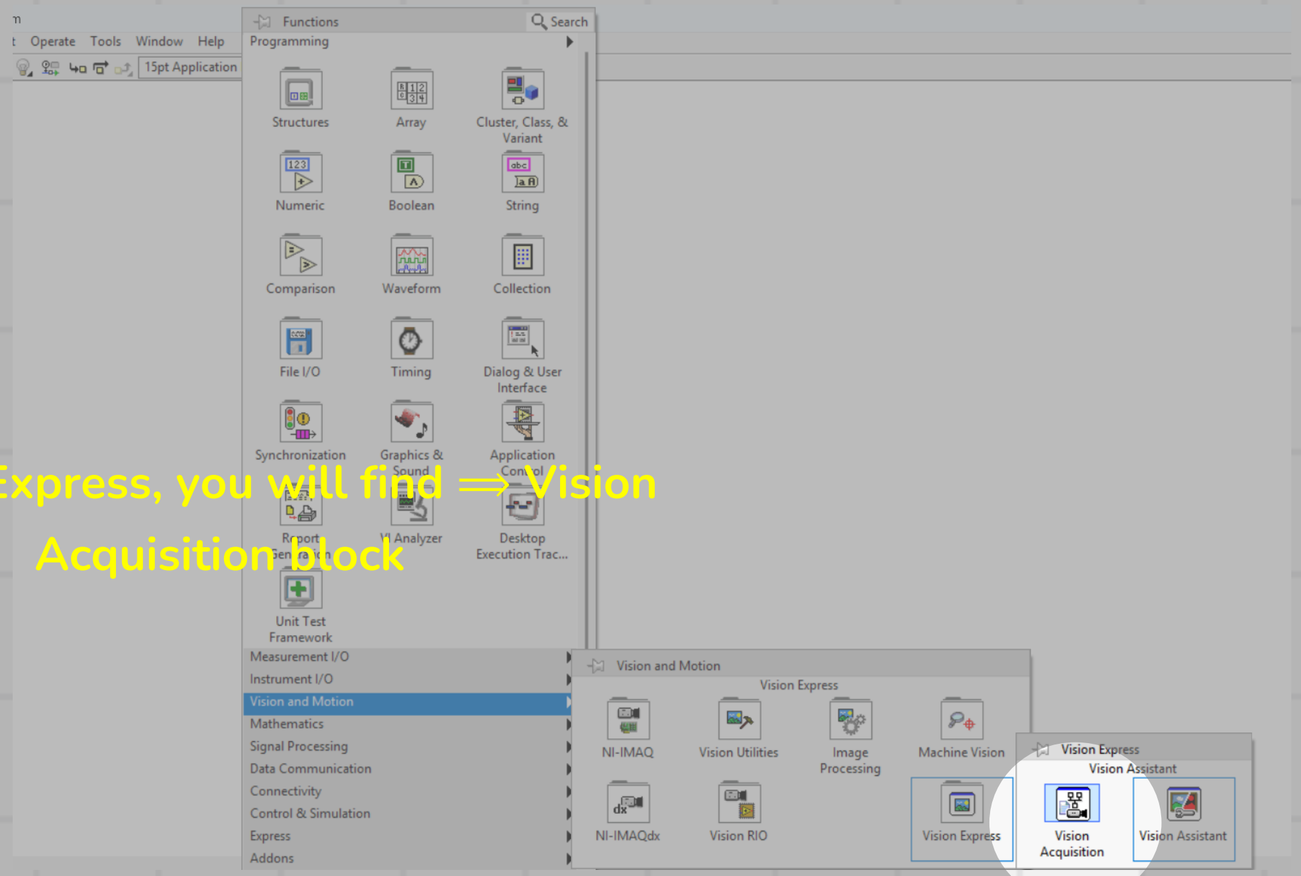


You must go ⟹ Vision and Motion list, then go to

⟹ Vision Express

Real-time Camera

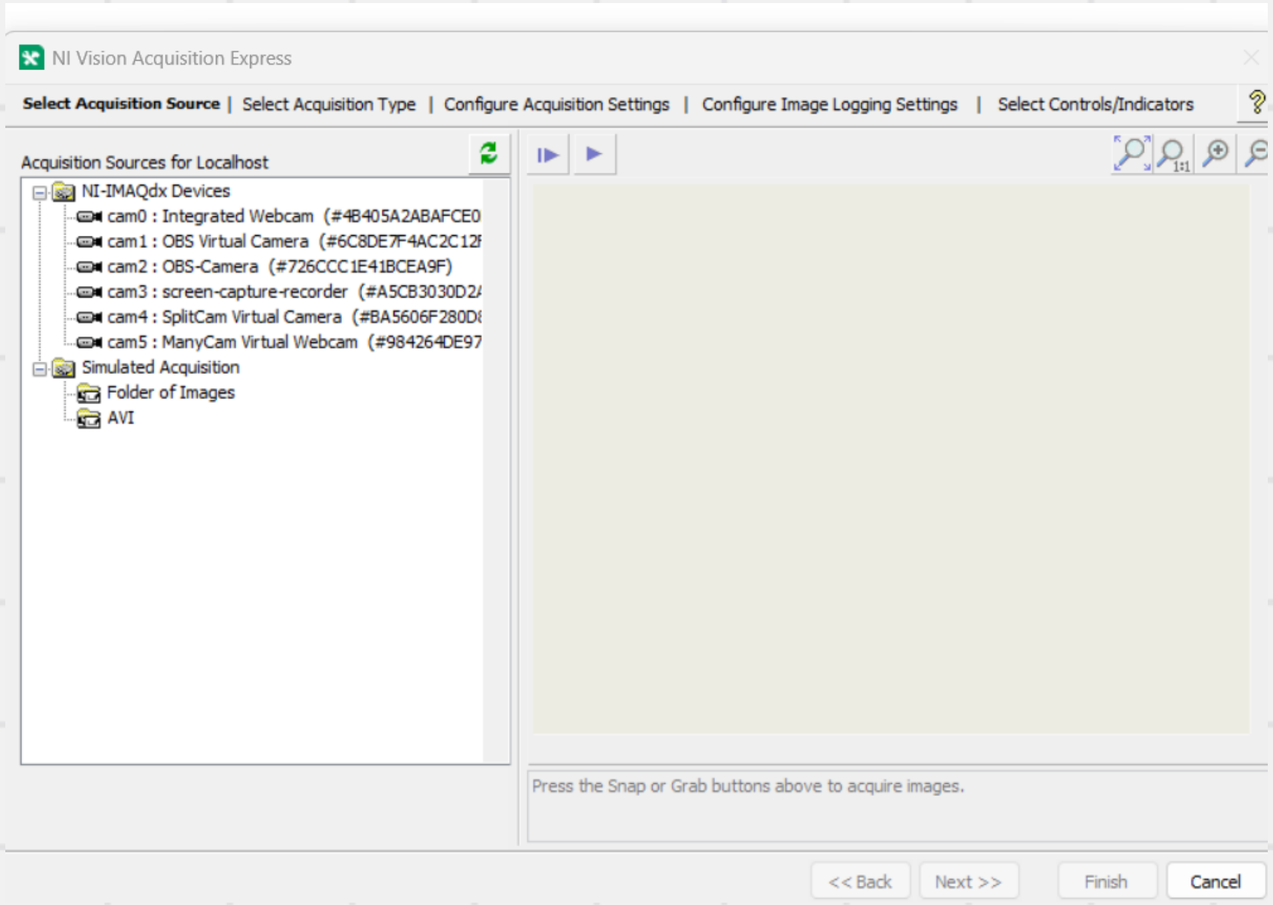
Simplest way to do this is by depending on vision acquisition app.



In Vision Express, you will find ⟹ Vision Acquisition block

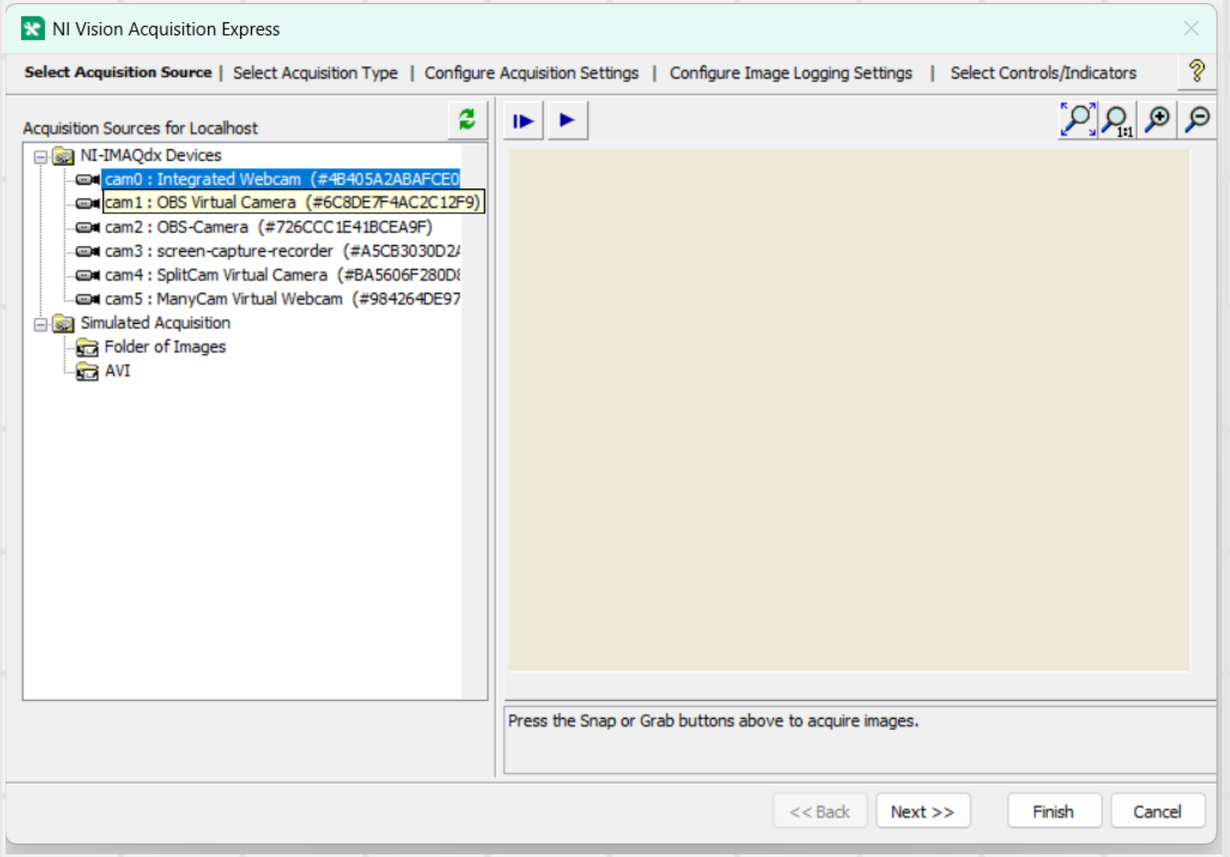
Real-time Camera

When you drag and drop this block, A GUI window will open.

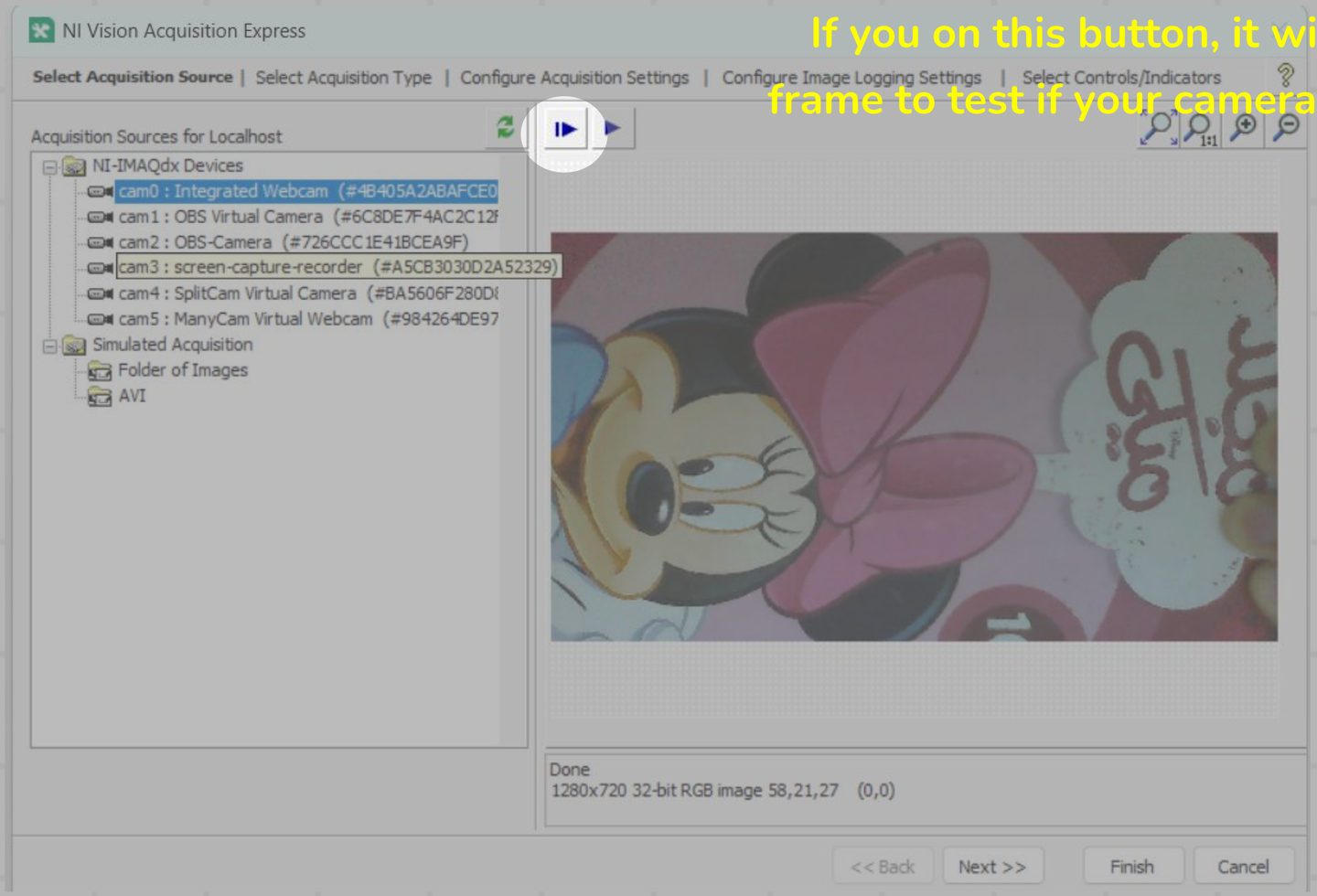


Real-time Camera

Now, You will find your connected cameras and choose the needed one.

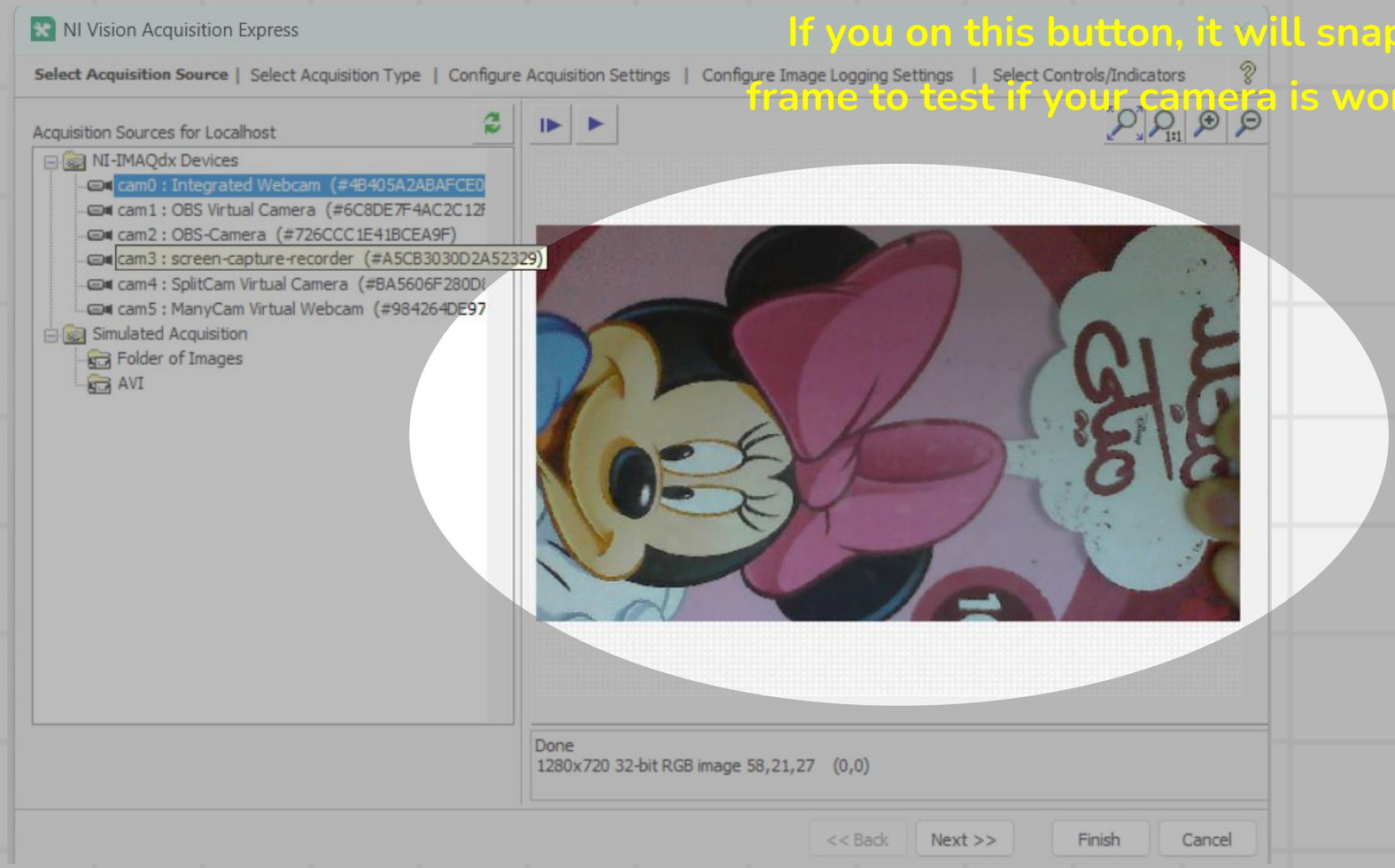


Real-time Camera



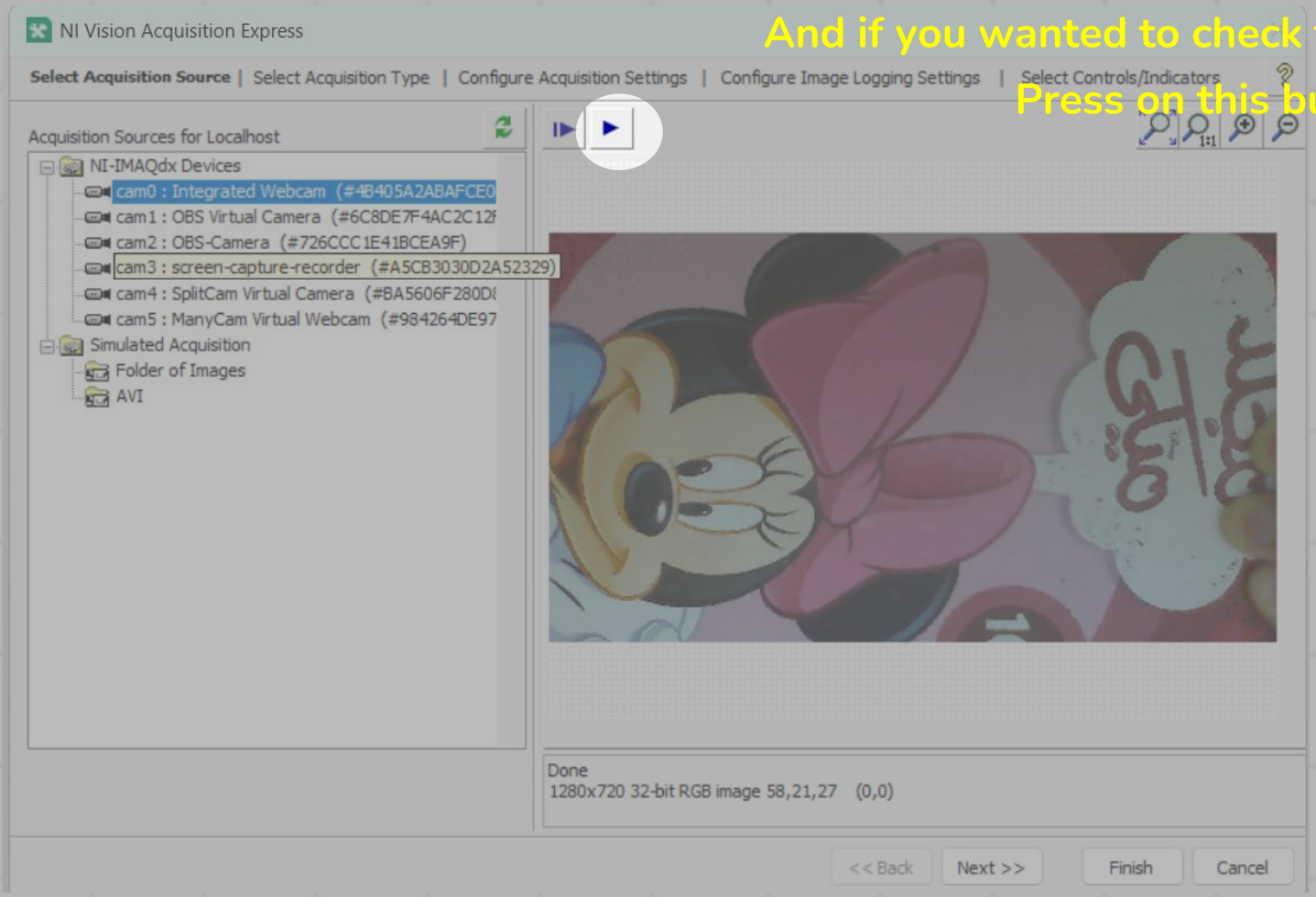
If you on this button, it will snap only one frame to test if your camera is working or not.

Real-time Camera



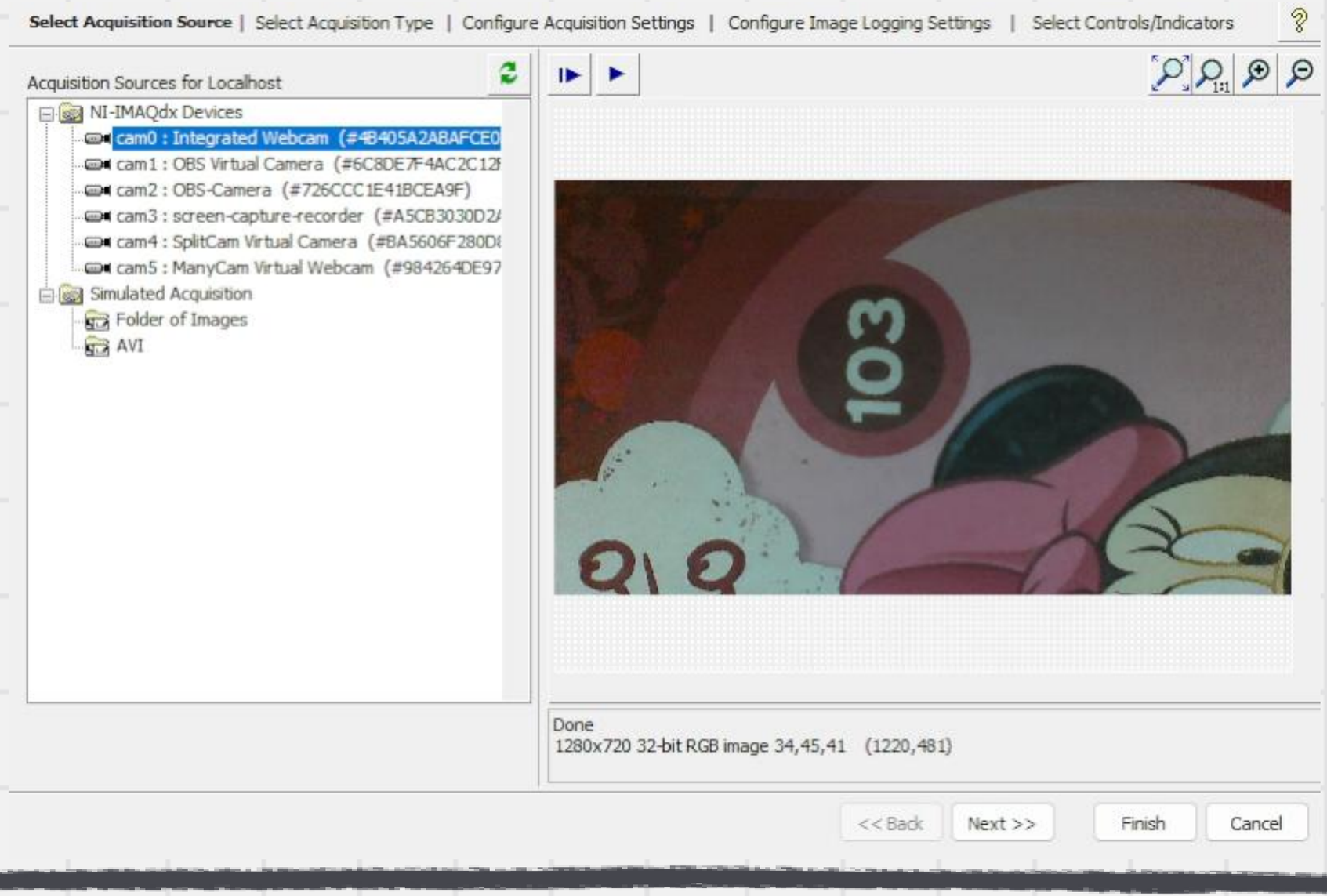
If you on this button, it will snap only one frame to test if your camera is working or not.

Real-time Camera



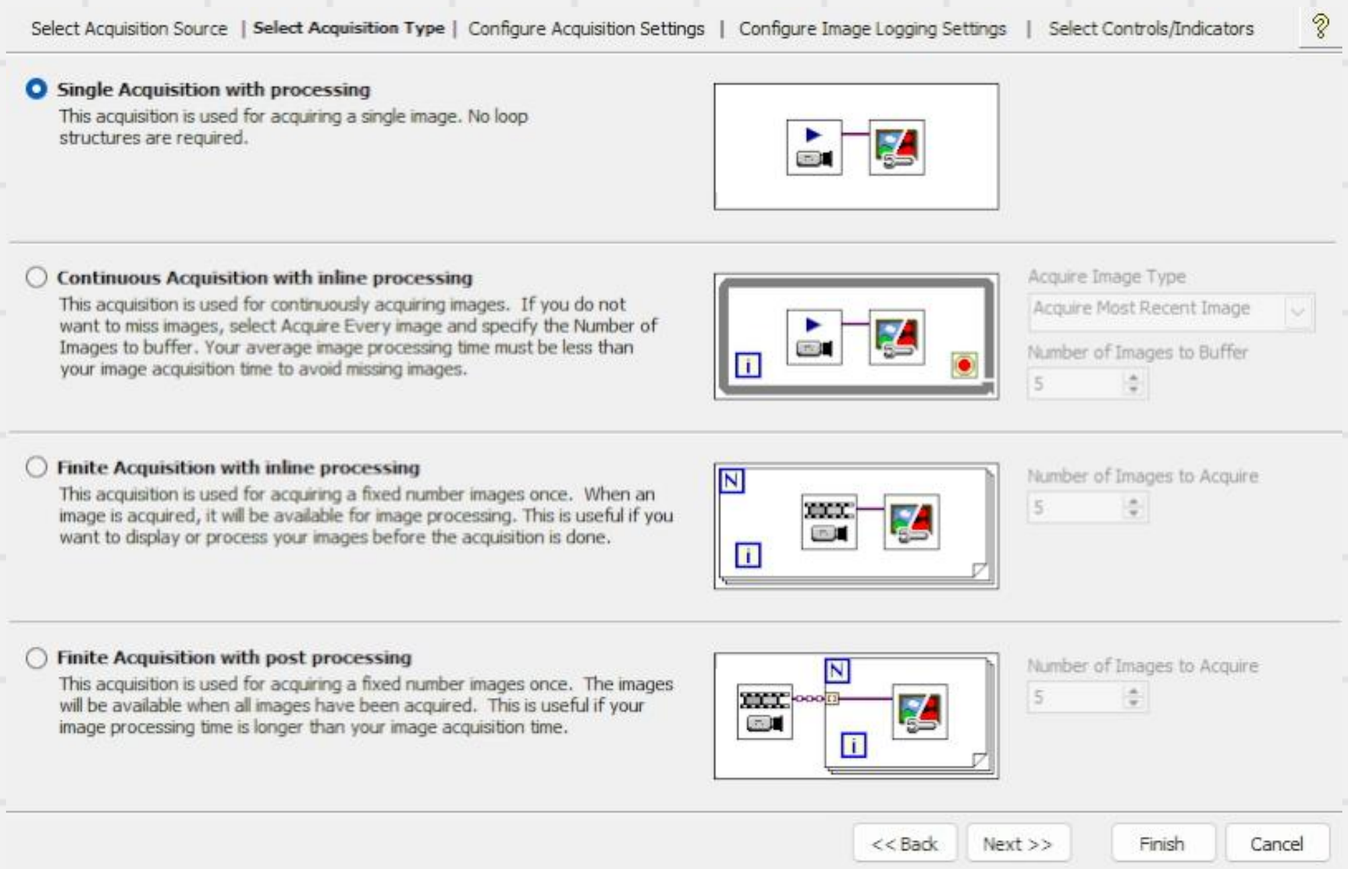
And if you wanted to check the real-time cam.
Press on this button.

Real-time Camera



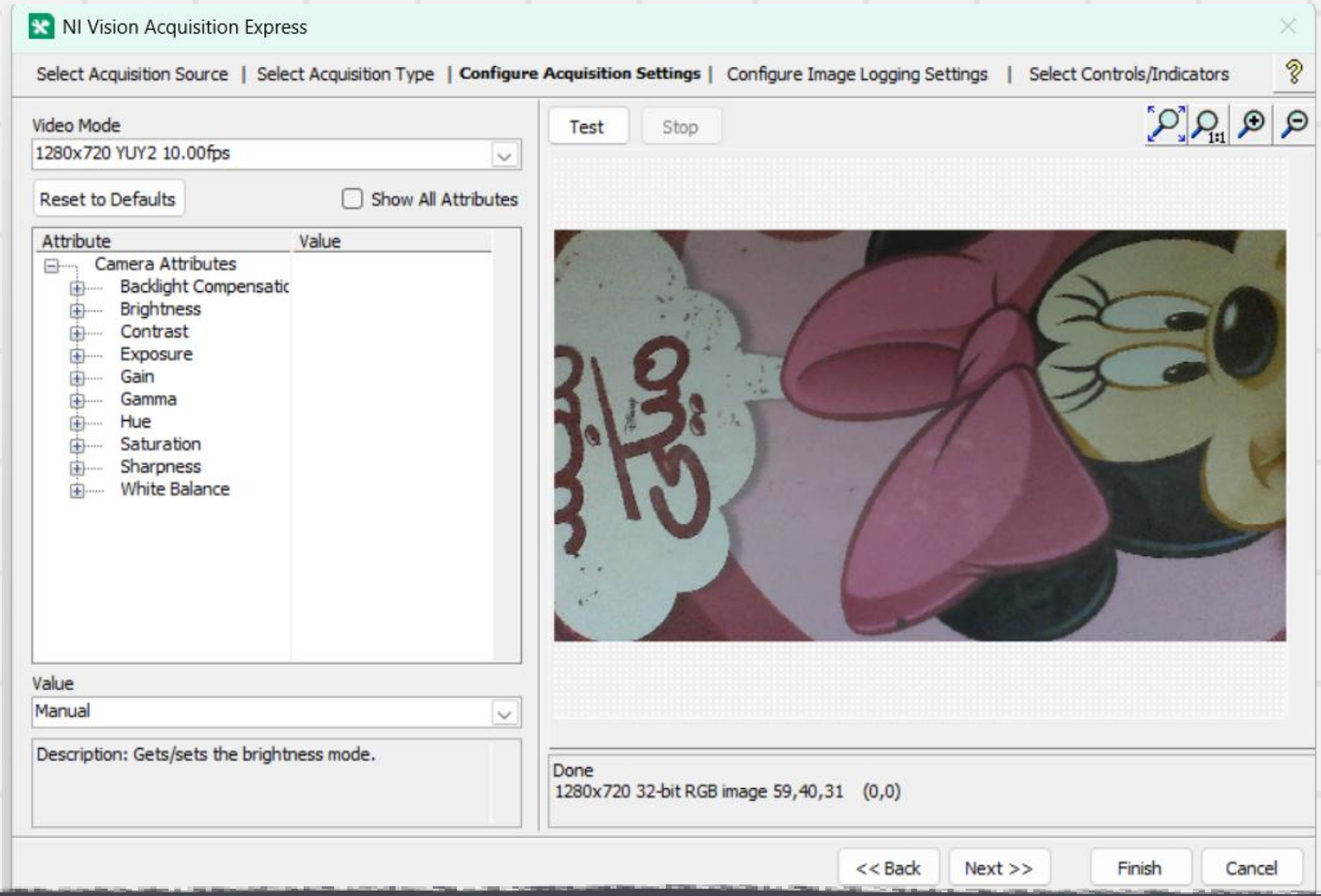
Real-time Camera

After pressing next, we will be able to select our suitable Acquisition type.



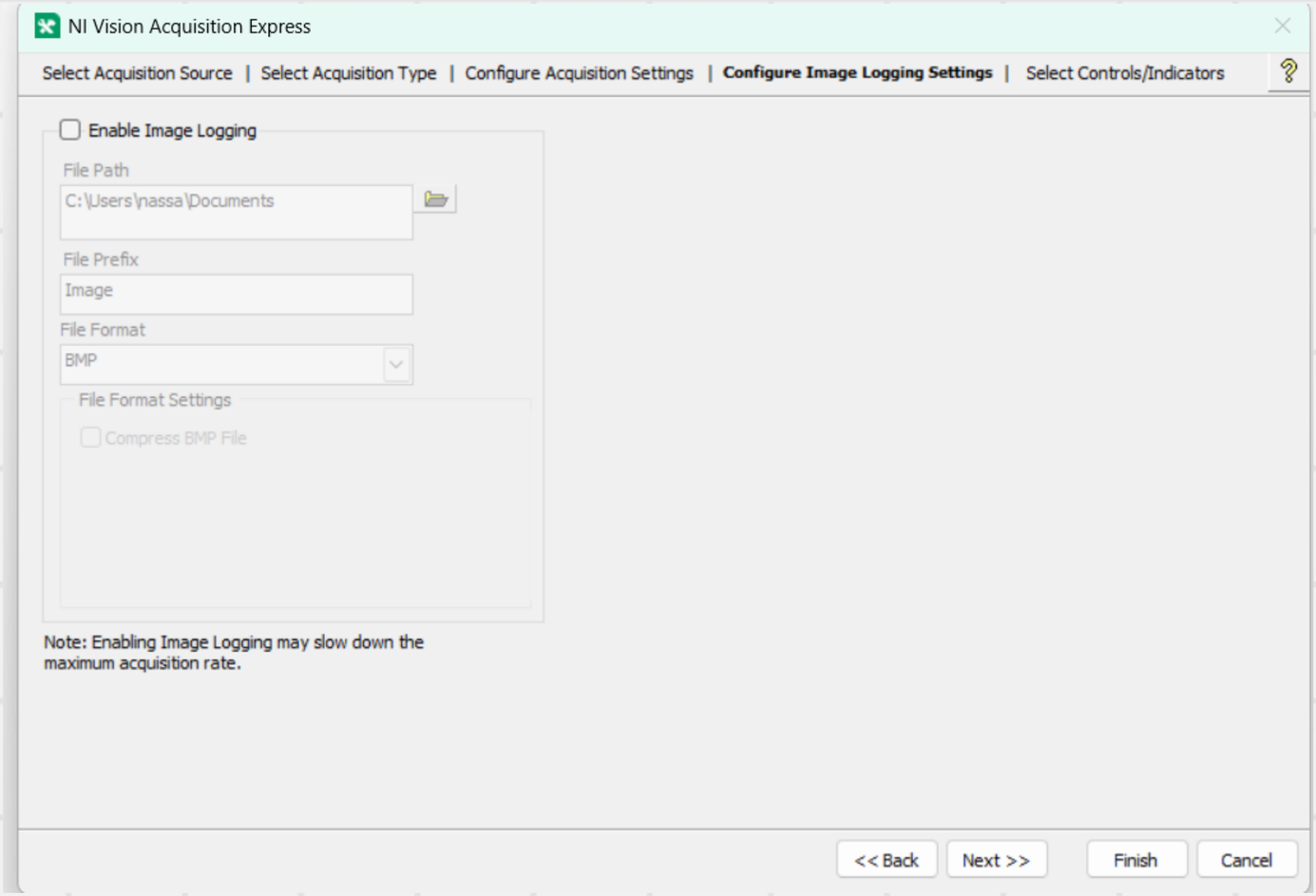
Real-time Camera

After that, we are going to choose the video mode which control the resolution and frame rate.



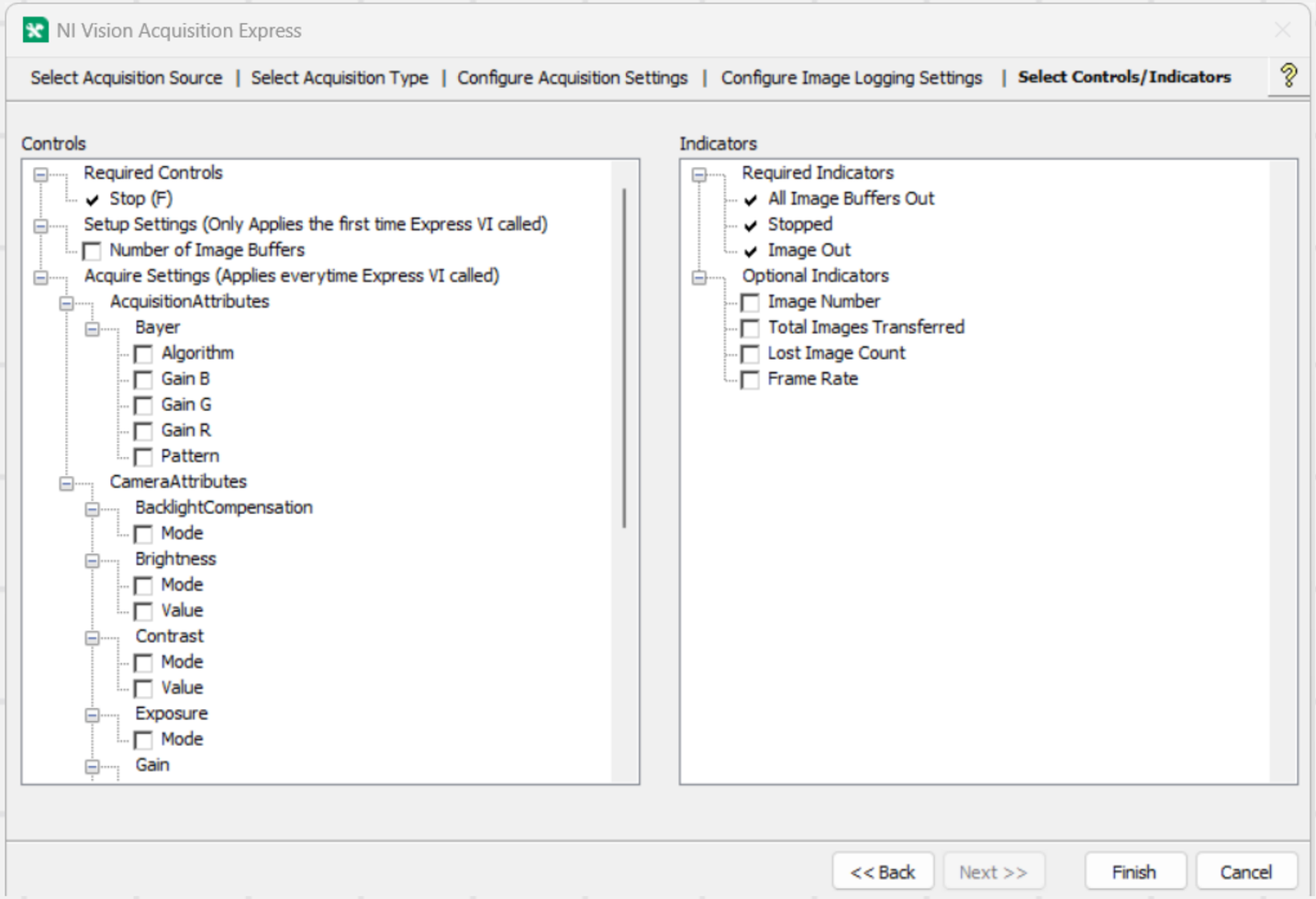
Real-time Camera

Now, it will offer you to enable image logging. For me I will neglect this option.



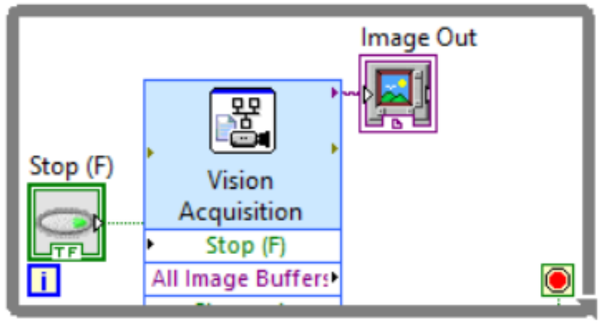
Real-time Camera

Finally, you will be able to enable controls and indicators.



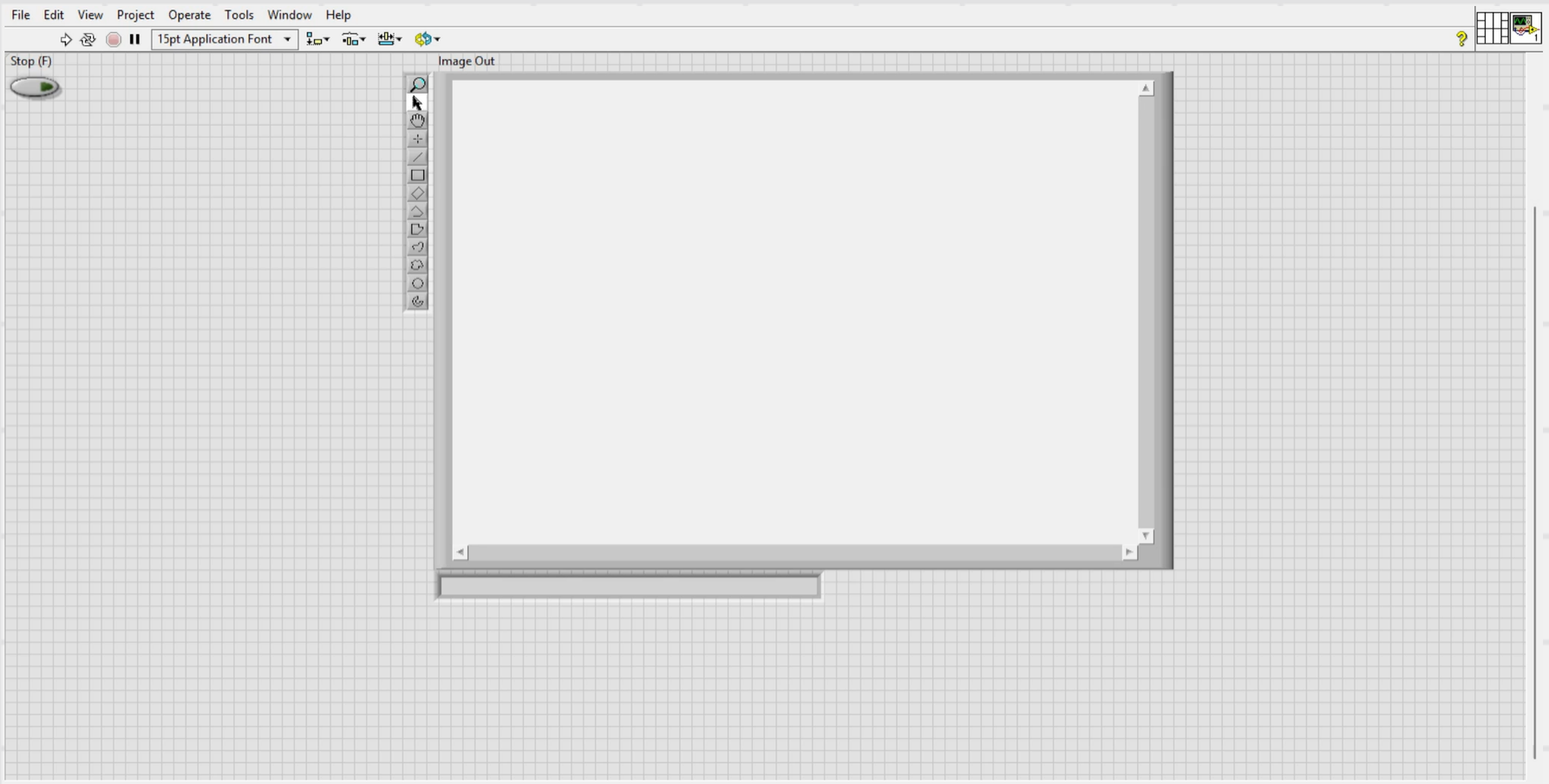
Real-time Camera

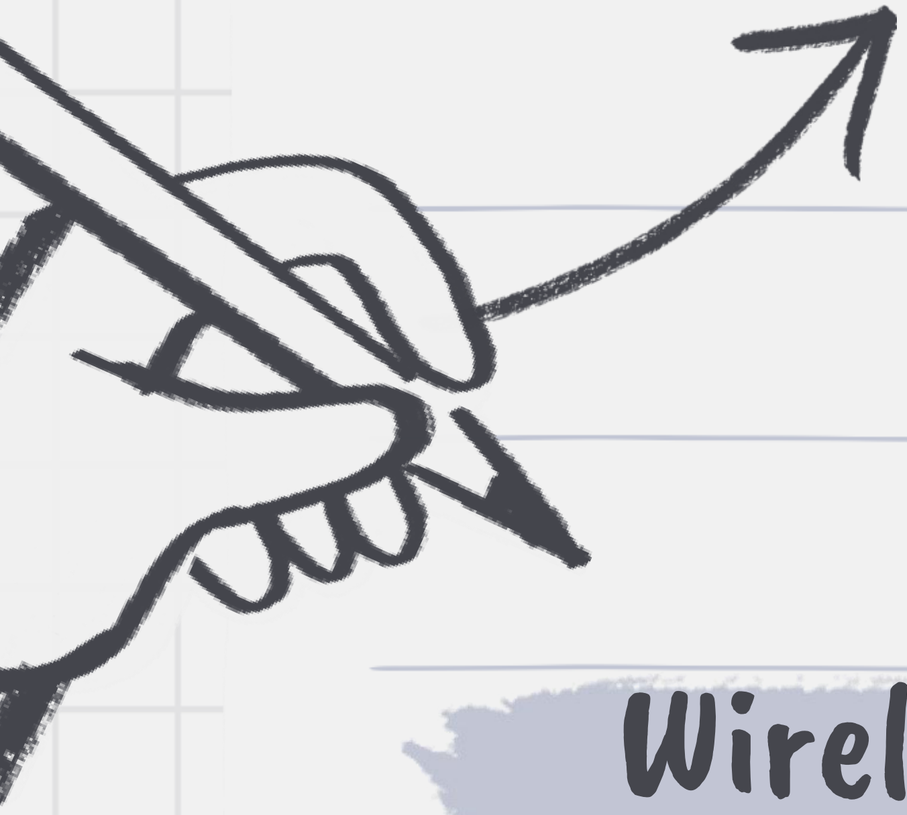
After pressing on finish, You will find that a while loop was generated in block diagram



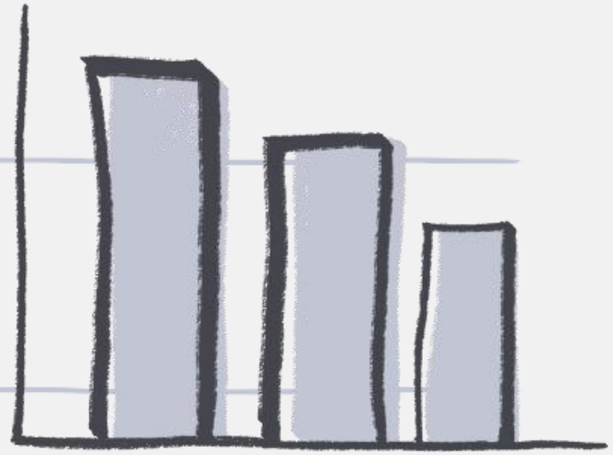
Real-time Camera

If you went to front panel, you will find Image out window.





03



Wireless Real-time Cam

Wireless Real-time Cam

In robotics and autonomous systems, accessing a real-time camera feed is essential for tasks like object tracking, navigation, and environment perception. While LabVIEW offers built-in support for USB and wired cameras, it does not directly support wireless IP cameras out of the box.

In this chapter, we'll explore how to overcome this limitation by integrating a wireless real-time camera—such as an IP camera—into LabVIEW. This allows us to stream video wirelessly, enabling more flexible and mobile robotic applications.

Step by step, we'll demonstrate how to receive the video stream, process it within LabVIEW, and use it as a foundation for computer vision tasks in robotics.

Wireless Real-time Cam

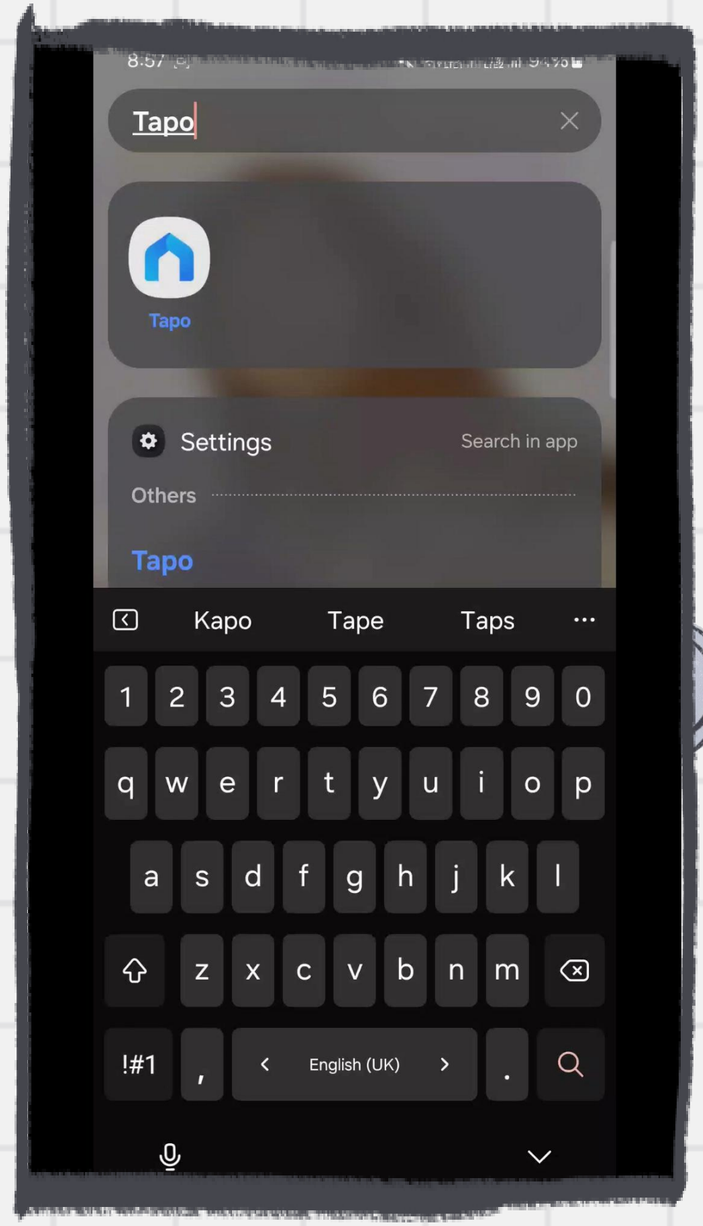
For this project, we use the TP-Link Tapo C100, a budget-friendly wireless IP camera that offers reliable video streaming over Wi-Fi. It supports up to 1080p resolution, night vision, and motion detection, making it a great choice for basic surveillance and computer vision applications.

While the Tapo C100 is not natively supported by LabVIEW, we can access its RTSP (Real-Time Streaming Protocol) video stream and integrate it into our LabVIEW environment using external tools. This allows us to use it as a wireless real-time camera, ideal for robotics and autonomous systems where mobility and flexibility are essential.

In this session, we'll show how to connect the Tapo C100 to LabVIEW and stream live video for further processing and object tracking.

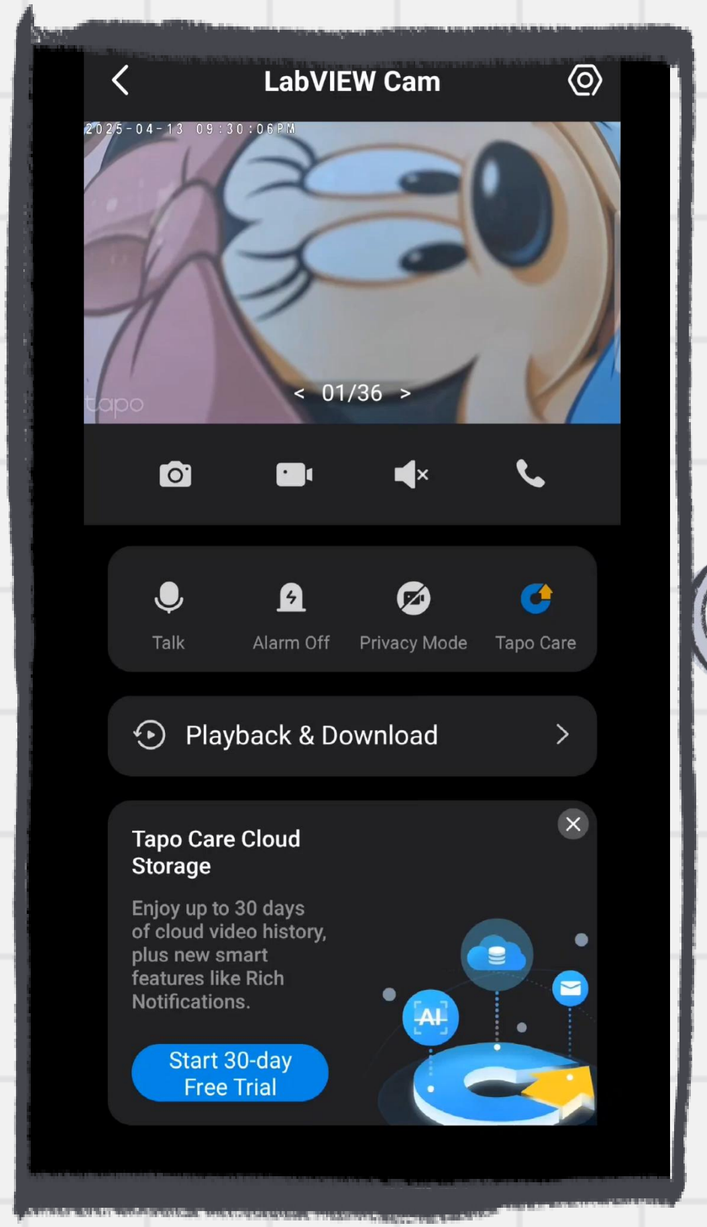
Wireless Real-time Cam

In this video, you will see how to start your camera’s stream.



Wireless Real-time Cam

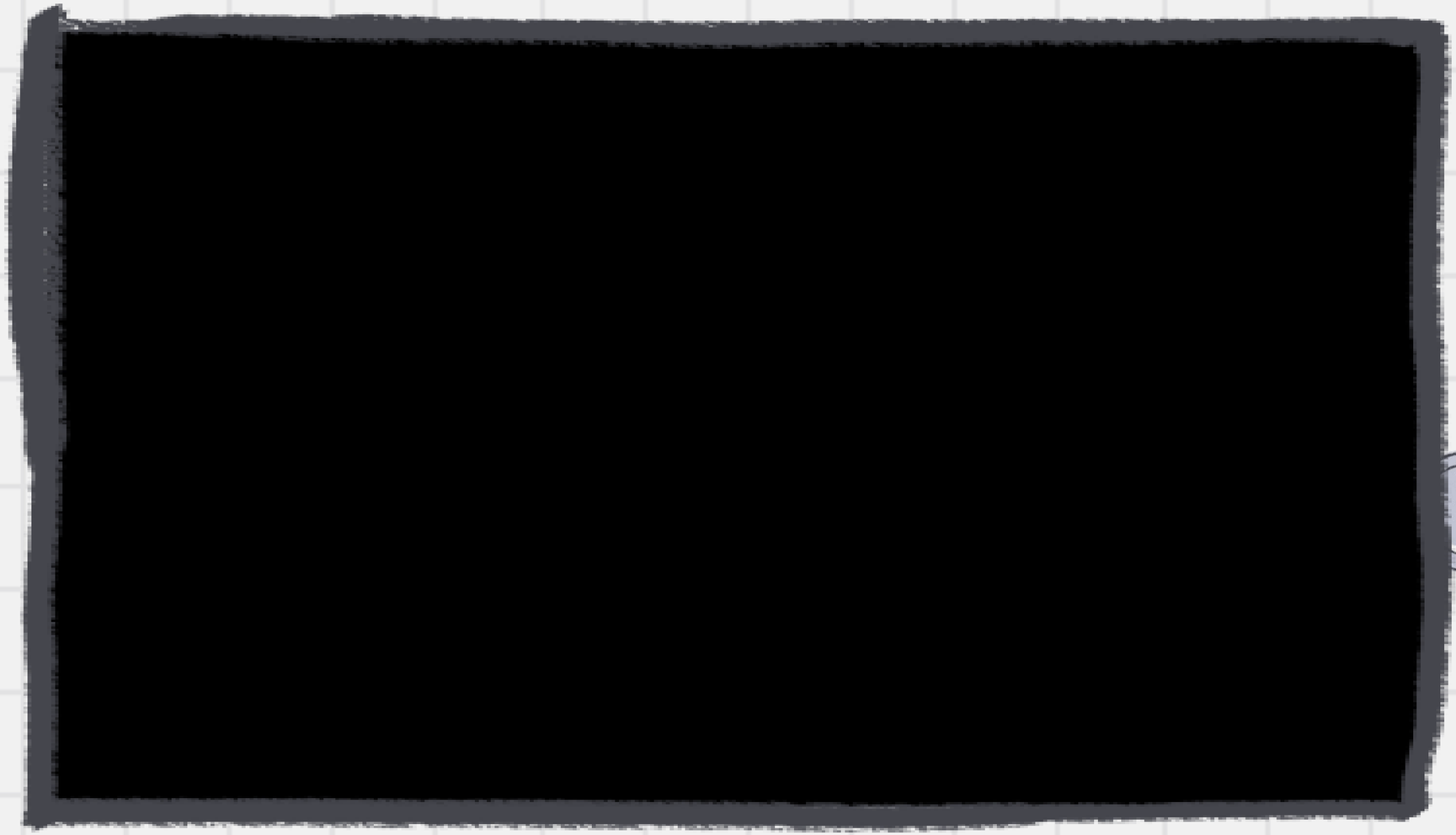
Now, we are ready to stream Tapo c100 video to our host machine.



Wireless Real-time Cam

Now, we will install app which can monitor the stream using the IP.
I tried many apps, and I found that SplitCam has the fastest response with no delay.

Wireless Real-time Cam



Add the following address to you IP camera input

“rtsp://$USER_NAME$:$PASSWORD$@192.168.xx.xx:554/stream1”

Wireless Real-time Cam

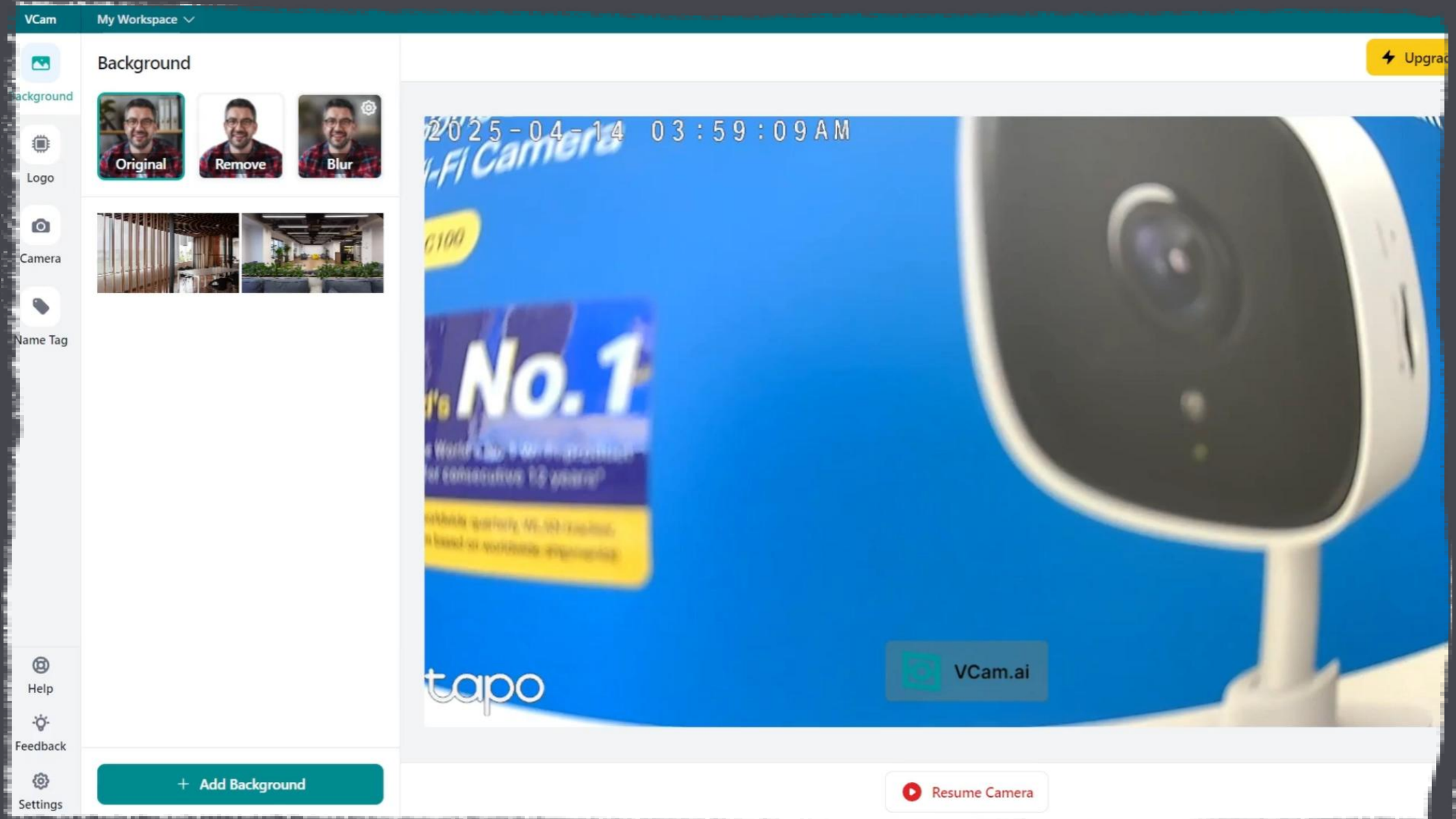
Now that the video stream is being published to our host machine, we face another challenge: how to use this stream as an input to LabVIEW.
As mentioned earlier, LabVIEW only supports image acquisition from wired cameras through drivers like Vision Acquisition. This means it doesn't natively accept video streams from wireless IP cameras.

So, how can we make this stream act like a wired camera?
The answer is simple: we use tools that create a virtual camera. A virtual camera appears to the system—and to LabVIEW—as if it were a real, wired webcam. However, its source can actually be an RTSP stream, a screen capture, or any other video input.

we'll show how to use virtual camera software to bridge the gap between the wireless Tapo C100 and LabVIEW, enabling real-time video processing for our robotics applications.

Wireless Real-time Cam

SplitCam offers virtual cam but for some reason it doesn’t work with LabVIEW, but I found app called VCam which worked seamlessly with LabVIEW. So, I take the SplitCam virtual cam as input in VCam, then I use VCam virtual cam as input at LabVIEW.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Background
Logo
Camera
Name Tag
Help
Feedback
Settings
Original
Remove
Blur
+ Add Background
Upgrad
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THANK
YOU